

# METHOD DESIGN DOCUMENT

## Superstore Sales Performance Analysis

Data Analytics Project

**Status** | Final Document

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**Deliverables** | Cleaned dataset, Python EDA notebook, Power BI dashboard, MDD

## 1. Project Overview

This Method Design Document (MDD) provides a comprehensive technical specification for the Superstore Sales Performance Analysis project. The project involves end-to-end data processing, exploratory data analysis (EDA), correlation analysis, and business intelligence (BI) visualization for a retail operations dataset.

Monitoring sales, profitability, and operational volume is critical in the retail industry. This project identifies the key product categories, regional performance metrics, and profitability drivers to translate raw transactional data into actionable business recommendations.

- **Project Name:** Superstore Sales Performance Analysis
- **Domain:** Retail / Sales Analytics
- **Type:** Data Analytics + Business Intelligence (BI)
- **Primary Tools:** Python 3 (Pandas, NumPy, Matplotlib, Seaborn), Power BI Desktop
- **Dataset Source:** SuperStore Sales DataSet.xlsx
- **Records:** 5,901 records across 23 initial features

### 1.1 Project Objectives

- Perform complete exploratory data analysis (EDA) on the Superstore sales dataset.
- Clean and pre-process raw data by handling missing values, dropping empty columns, and correcting data types.
- Engineer new temporal and financial features to enable deeper profitability analysis.
- Identify key sales trends across years, regions, and product sub-categories through visual analysis.
- Build an interactive Power BI dashboard for operational monitoring using the Clean\_Superstores\_sales.xlsx file.
- Document the complete design, methodology, and findings.
- **Implement Predictive Analytics:** Utilize Power BI's built-in forecasting capabilities to generate a 15-day future sales trend based on historical transaction data, allowing for proactive inventory and revenue planning.

### 1.2 Scope

- **Dataset:** Retail operations dataset containing 5,901 clean records with 21 final features.
- **Analysis type:** Exploratory Data Analysis + Correlation Analysis.
- **Deliverables:** Clean dataset, Python notebook, Power BI dashboard (.pbix), MDD.

## 2. Dataset Description

## 2.1 Source

The dataset used for this project is the SuperStore Sales DataSet.xlsx, containing real-world retail transaction records.

## 2.2 Dataset Summary

- **Total Records (raw):** 5,901 rows
- **Total Columns (raw):** 23 columns
- **Missing Values (post-cleaning):** 0
- **Duplicate Records:** 0
- **Target Variables:** Sales and Profit

## 2.3 Column Descriptions

| Column Name                     | Data Type | Description                                                 |
|---------------------------------|-----------|-------------------------------------------------------------|
| Row ID+O6G3A1:R6                | Int64     | Unique identifier for each row                              |
| Order ID                        | Object    | Unique identifier for each customer order                   |
| Order Date                      | DateTime  | Date the order was placed                                   |
| Ship Date                       | DateTime  | Date the order was shipped                                  |
| Ship Mode                       | Object    | Shipping classification (e.g., Standard Class, First Class) |
| Customer ID                     | Object    | Unique identifier for each customer                         |
| Customer Name                   | Object    | Name of the purchasing customer                             |
| Segment                         | Object    | Customer segment (e.g., Consumer, Corporate)                |
| Country / City / State / Region | Object    | Geographic details of the order destination                 |

|                                  |         |                                                |
|----------------------------------|---------|------------------------------------------------|
| <b>Product ID / Product Name</b> | Object  | Identifier and name of the purchased item      |
| <b>Category / Sub-Category</b>   | Object  | High-level and detailed product classification |
| <b>Sales</b>                     | Float64 | Total sales value of the transaction           |
| <b>Quantity</b>                  | Int64   | Number of items purchased in the transaction   |
| <b>Profit</b>                    | Float64 | Net profit generated from the transaction      |
| <b>Returns</b>                   | Float64 | Indicator of whether an item was returned      |
| <b>Payment Mode</b>              | Object  | Method of payment used (e.g., Online, Cards)   |

(Note: ind1 and ind2 were present in the raw data but contained no valid data).

### 3. Methodology

The project followed a structured analytics pipeline from raw data ingestion through to interactive dashboard delivery. All data processing and EDA were performed using Python, and the cleaned dataset was imported into Power BI Desktop for dashboard creation.

| Layer                         | Component                                           | Technology           | Output              |
|-------------------------------|-----------------------------------------------------|----------------------|---------------------|
| <b>1. Ingestion</b>           | Raw Excel Loading                                   | Python / Pandas      | DataFrame           |
| <b>2. Cleaning</b>            | Missing values, column dropping, duplication checks | Python / Pandas      | Cleaned DataFrame   |
| <b>3. Feature Engineering</b> | Date extraction, Profit Margin calculation          | Python / Pandas      | Augmented DataFrame |
| <b>4. EDA</b>                 | Statistical analysis                                | Seaborn / Matplotlib | Charts & Insights   |

|                                |                                     |                  |                    |
|--------------------------------|-------------------------------------|------------------|--------------------|
|                                | and visualization                   |                  |                    |
| <b>5. Statistical Analysis</b> | Cross-variable correlation analysis | Seaborn / Pandas | Correlation Matrix |
| <b>6. Power BI Dashboard</b>   | Interactive visual analytics        | Power BI Desktop | PBIX Dashboard     |

## 4. Data Pre-processing

The raw dataset was inspected for quality issues before any analysis. The following pre-processing steps were applied sequentially in the Python Notebook.

### Step 1 - Data Loading

The Excel file was loaded using `pd.read_excel()`, producing a DataFrame with 5,901 rows and 23 columns. The initial structure was inspected using `data.head()` and `data.info()`.

### Step 2 - Missing Value Detection & Handling

A missing value check (`data.isnull().sum()`) revealed three columns with null values:

- **Returns:** 5,614 missing values. These were filled with 0 using `fillna(0)`, establishing a clean numeric indicator for returned items.
- **ind1 and ind2:** Both columns contained 5,901 missing values (100% null). These columns were entirely dropped from the dataset using `drop(columns=['ind1','ind2'])`.

### Step 3 - Duplicate Detection

The dataset was checked for exact row matches using `data.duplicated()`. No duplicate records were found, ensuring data integrity.

### Step 4 - Feature Engineering & Data Type Correction

- **Date Conversion:** Order Date was strictly converted to DateTime format using `pd.to_datetime()`.
- **Temporal Features:** New columns for Year and Month were extracted from the Order Date feature to enable time-series aggregation.
- **Financial Features:** A Profit Margin (%) feature was engineered by calculating  $(\text{Profit} / \text{Sales}) * 100$ . This enables percentage-based profitability tracking irrespective of raw sales volume.

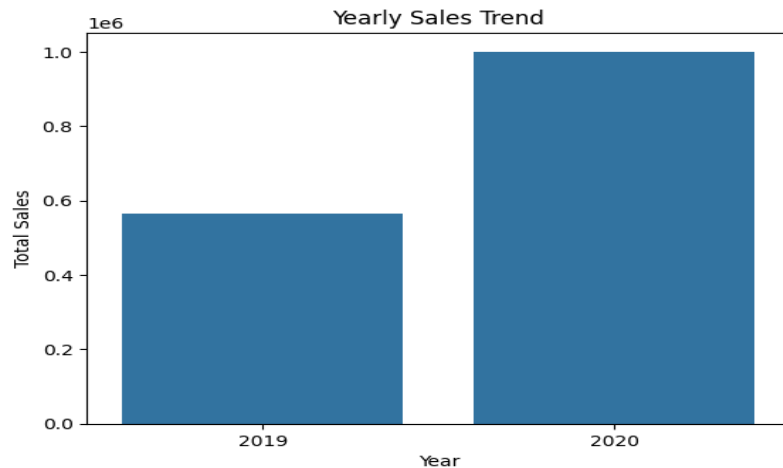
### Step 5 - Export Cleaned Data

The finalized dataset was prepared for export. This file, **Clean\_Superstores\_sales.xlsx**, serves as the sole data source for the Power BI dashboard, ensuring full consistency between Python analysis and the BI layer.

## 5. Exploratory Data Analysis (EDA)

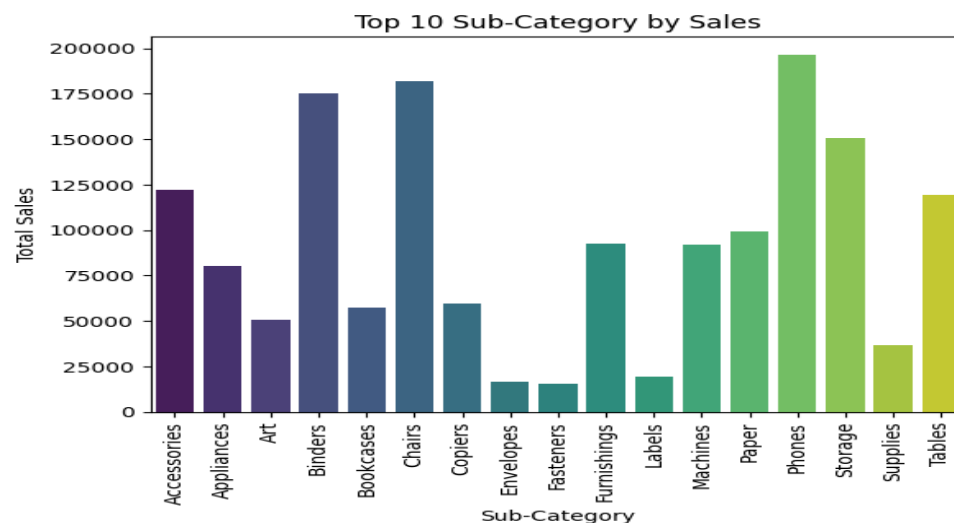
EDA was conducted using Matplotlib and Seaborn to identify financial patterns and categorical relationships.

### 5.1 Yearly Sales Trend



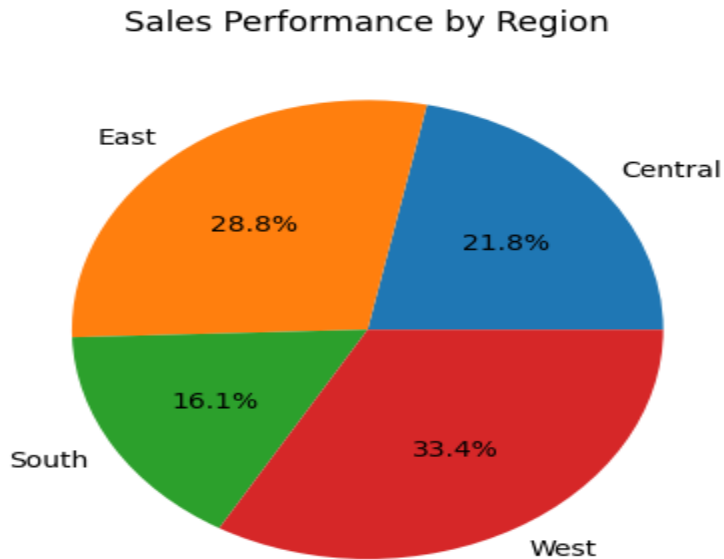
A bar chart was generated to visualize the sum of sales aggregated by Year. This tracks the macroeconomic growth or contraction of the Superstore's total revenue over the observed timeframe.

### 5.2 Sub-Category Sales Performance



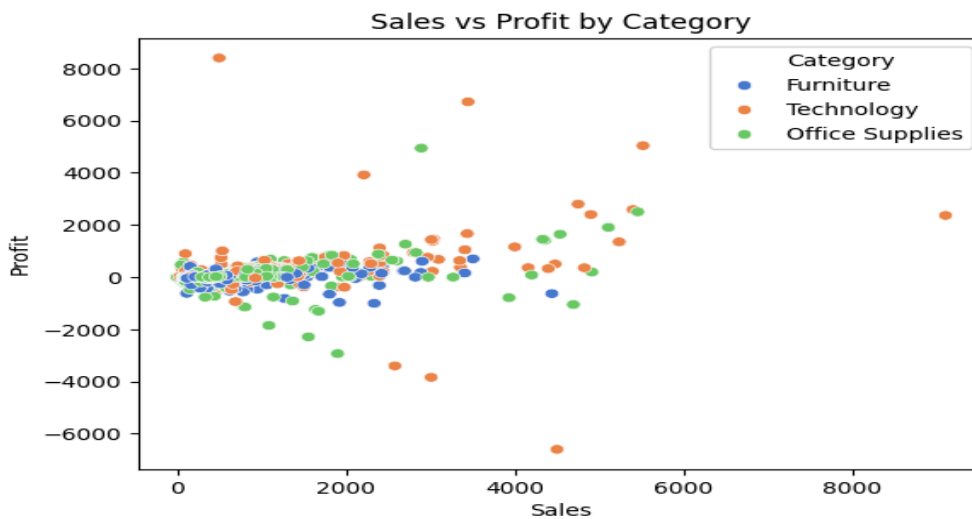
A bar plot displaying the Top 10 Sub-Categories by Sales was created using a viridis color palette. This highlights the specific product lines driving the highest revenue volume.

## 5.3 Regional Sales Distribution



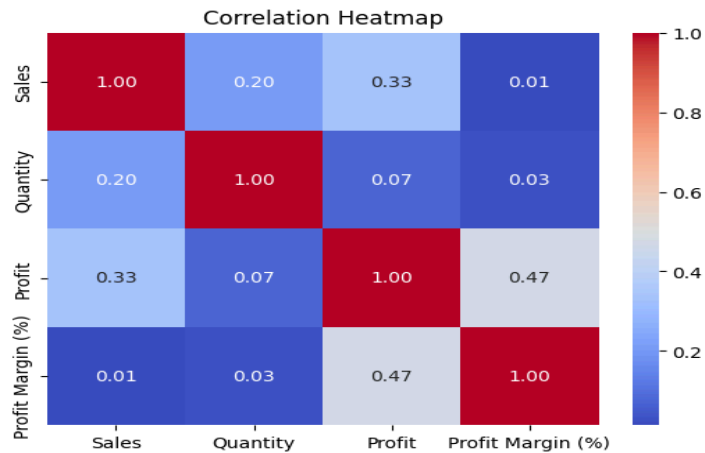
A pie chart was utilized to analyze Sales Performance by Region, with percentage labels automatically mapped. This determines geographic market share and identifies the most lucrative retail territories.

## 5.4 Profitability by Category (Sales vs. Profit)



A scatter plot was engineered mapping Sales on the X-axis against Profit on the Y-axis, separated by Category using color hues. This chart is highly critical as it visually separates high-volume/high-profit transactions from high-volume/loss-making transactions.

## 6. Correlation Analysis

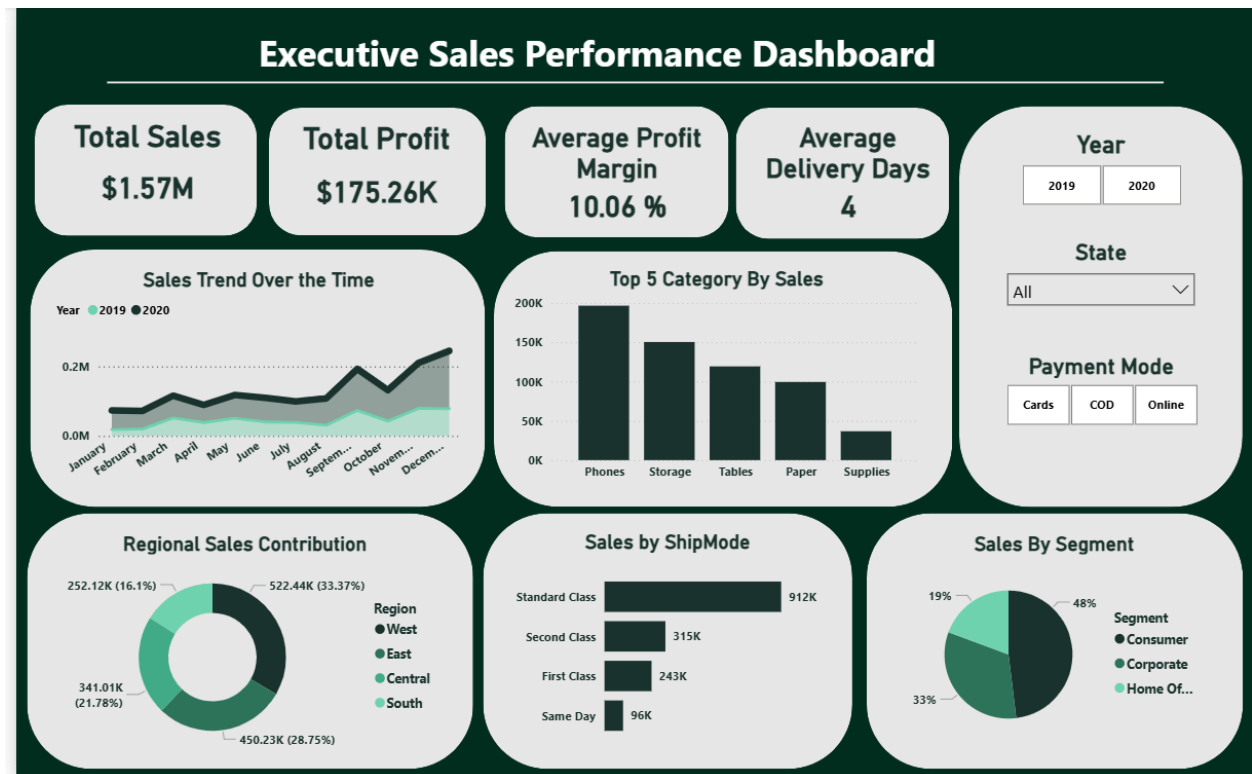


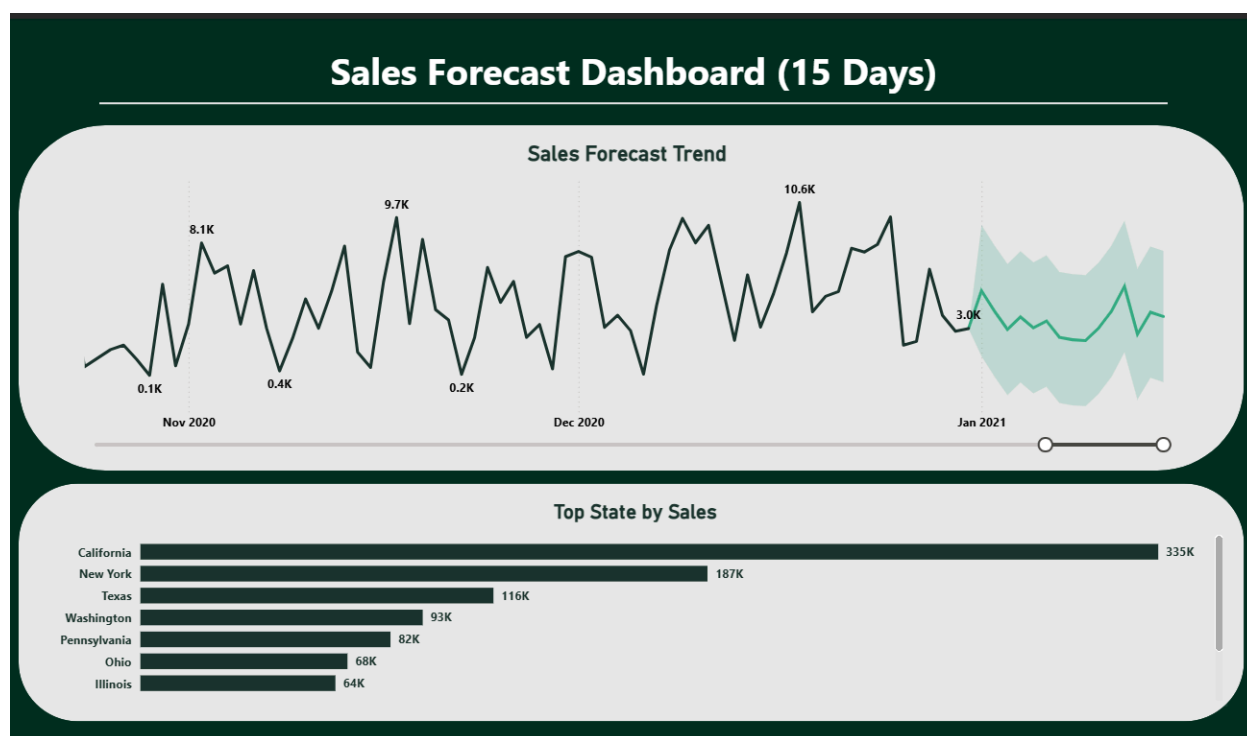
To quantify the linear relationships between numeric financial metrics, a correlation matrix was computed and visualized via a Seaborn heatmap.

**Analyzed Features:** Sales, Quantity, Profit, and Profit Margin (%).

This statistical check ensures that we understand the baseline behavior of the data—for instance, verifying if higher quantities inherently drive higher profit, or if discounting (affecting profit margin) inversely correlates with total sales volume.

## 7. Power BI Dashboard





## 7.1 Overview

The **Clean\_Superstores\_sales.xlsx** dataset was imported into Power BI Desktop to build an interactive operational dashboard. The dashboard translates the Python EDA findings into a dynamic environment where stakeholders can filter by date, region, and product category.

## 7.2 Dashboard Components (Analytical Mapping)

| Visual Type | Data Fields Used                  | Business Insight                              |
|-------------|-----------------------------------|-----------------------------------------------|
| KPI Card    | Sales (SUM)                       | Total aggregate revenue tracking.             |
| KPI Card    | Profit (SUM)                      | Overall net profit generated.                 |
| KPI Card    | Profit Margin (%)<br>(AVERAGE)    | Average transaction health and profitability. |
| Line Chart  | Order Date (Year/Month),<br>Sales | Dynamic temporal trend of revenue.            |
| Donut Chart | Region, Sales                     | Geographic distribution of                    |



|                                 |                         |                                                                                                                                                                                                     |
|---------------------------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 |                         | total revenue.                                                                                                                                                                                      |
| <b>Clustered Bar Chart</b>      | Sub-Category, Profit    | Identifying loss-leading vs. highly profitable product lines.                                                                                                                                       |
| <b>Slicers</b>                  | Year, Category, Segment | Enables interactive, granular drill-downs for end-users.                                                                                                                                            |
| <b>Line Chart with Forecast</b> | Order Date, Sales       | <b>15-Day Sales Forecast:</b><br>Analyzes historical time-series data to predict revenue volume for the next 15 days, complete with upper and lower confidence bounds for accurate demand planning. |

## 8. Business Recommendations

Based on the data processing and visual analysis pipeline, the following strategic actions are recommended:

1. **Margin Optimization:** Utilize the Profit Margin (%) and Sales vs. Profit scatter plot to identify product sub-categories that generate high sales but negative net profit. Review discount strategies or supplier costs for these specific items.
2. **Regional Resource Allocation:** Reallocate marketing budgets and supply chain resources based on the regional sales distribution pie chart, ensuring high-performing regions maintain stock availability.
3. **Temporal Planning:** Leverage the Yearly and Monthly sales trends to forecast peak seasons. Increase inventory and staffing immediately preceding historically high-sales months.
4. **Proactive Inventory Planning (15-Day Forecast):** The dashboard's 15-day sales forecast provides a direct line of sight into upcoming demand. Store managers should align this 15-day predictive window with supply chain lead times to ensure high-velocity items are restocked *before* the forecasted sales spikes occur.

## 9. Tools & Technologies

| Tool / Library       | Purpose                                                                            |
|----------------------|------------------------------------------------------------------------------------|
| Python 3             | Core programming language for all data processing.                                 |
| Pandas & NumPy       | Data ingestion, missing value imputation, feature engineering (Profit Margin (%)). |
| Matplotlib & Seaborn | Statistical visualization, scatter plots, correlation heatmaps.                    |
| Power BI Desktop     | Interactive dashboard creation, KPI tracking, dynamic filtering.                   |

## 10. Limitations & Future Work

- **Advanced Machine Learning Forecasting:** While a 15-day sales forecast is successfully implemented within the Power BI layer, future work could bring this predictive modeling directly into the Python pipeline using advanced Machine Learning algorithms (like ARIMA, Prophet, or XGBoost) to predict profit margins and item-specific demand over much longer horizons (e.g., 6–12 months).
- **Discount Impact:** The dataset lacks a dedicated 'Discount' feature. Incorporating discount percentages would allow for a deeper diagnosis of why certain high-sales transactions result in negative profit margins.
- **Customer Lifetime Value (CLV):** Future work could group by Customer ID to model retention, purchase frequency, and overall customer lifetime value.